Angular Resolution in the Partition Algebra: Diffraction Limits as Partition Theorems, Spherical-Harmonic Decomposition, and Calorimeter Granularity at the LHC

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Abstract

We prove that the angular resolution limit of any partition-structured detector is determined by the partition angular quantum number ℓ and is given by $\Delta\theta_\ell = \pi/(2\ell_{\max}+1)$, where $\ell_{\max} \leq n-1$ is the maximum angular sublevel at partition depth n (Theorem 3.2). This is the partition-algebraic derivation of the Rayleigh criterion $\Delta\theta \sim \lambda/D$ without invoking wave optics: the diffraction limit is a counting theorem, not a wave theorem.

The $2(2\ell+1)$ degeneracy of each partition sublevel provides the natural basis for spherical-harmonic decomposition of detector signals: the $2\ell+1$ azimuthal sub-states $m \in \{-\ell, \ldots, +\ell\}$ correspond to independent angular readout channels, and the spherical harmonics $Y_\ell^m(\theta, \phi)$ are the eigenstates of the partition angular operators (Theorem 2.2).

We prove the Angular Sampling theorem: a detector with calorimeter granularity $\Delta\eta \times \Delta\phi$ can resolve all angular modes up to $\ell_{\max} = \pi/\Delta\theta - \frac{1}{2}$, beyond which aliasing occurs (Theorem 4.1). For ATLAS EM calorimeter granularity $\Delta\eta \times \Delta\phi = 0.025 \times 0.025$, the maximum resolvable angular mode is $\ell_{\max} \approx 125$.

We derive the angular composition-inflation formula: the number of distinguishable angular patterns at partition level n with d detector branches is $A(n, d) = (2n-1)d(d+1)^{n-1}$, growing faster than the state count $T(n, d)$ by a factor of $(2n-1)$.

Keywords: angular resolution; partition algebra; diffraction limit; Rayleigh criterion; spherical harmonics; calorimeter granularity; angular sampling; composition-inflation; $SO(3)$

Contents

1 Introduction

The Rayleigh angular resolution criterion $\Delta\theta = 1.22\lambda/D$ is standardly derived from the first zero of the Airy function, which is a consequence of Fraunhofer diffraction of a plane wave through a circular aperture. This derivation requires wave optics, Fourier analysis, and the assumption of coherent illumination.

We show that the Rayleigh criterion is, at its core, a counting theorem: it expresses the impossibility of distinguishing two angular modes whose separation is smaller than the angular spacing of adjacent partition ℓ levels. The wave-optics derivation is one realisation of this counting theorem; the partition algebra provides another, valid at particle energies where wave optics does not apply.

2 The Partition Angular Structure

2.1 Angular quantum numbers as partition coordinates

From the partition coordinate system (n, ℓ, m, s) established in Sachikonye (a), the angular structure at principal level n consists of:

- $\ell \in \{0, 1, \ldots, n-1\}$: the angular sublevel (polar).
- $m \in \{-\ell, -\ell+1, \ldots, +\ell\}$: the azimuthal projection.
- Degeneracy at each ℓ: $2\ell+1$ states for each sublevel.
- Total angular states at level n: $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$.

Theorem 2.1 (Angular Degeneracy and Partition Capacity)**.** *The total number of angular states at partition level n is n^2, and the full partition capacity including spin is $C(n) = 2n^2$. The factor of 2 is the spin degeneracy $2s+1 = 2$ for $s = \frac{1}{2}$.*

Proof. $\sum_{\ell=0}^{n-1}(2\ell+1) = n + 2\sum_{\ell=1}^{n-1}\ell = n + 2 \cdot \frac{(n-1)n}{2} = n + n(n-1) = n^2$. With spin: $2n^2 = C(n)$, consistent with Sachikonye (a). □

2.2 Spherical harmonics as partition eigenstates

Theorem 2.2 (Partition Angular Eigenstates)**.** *The partition states $|n, \ell, m\rangle$ with fixed n and varying (ℓ, m) span the same Hilbert space as the spherical harmonics $\{Y_\ell^m(\theta, \phi)\}$. Specifically, the partition angular operator $\hat{L}^2$ has eigenvalues $\hbar^2\ell(\ell+1)$ and $\hat{L}_z$ has eigenvalues $\hbar m$, identical to the quantum mechanical angular momentum operators in the $SO(3)$ representation.*

Proof. The partition Lagrangian of Sachikonye (b) generates, via its angular Euler-Lagrange equations, the equation of motion: $\ddot{\theta} = -\Gamma^\theta_{\phi\phi}\dot{\phi}^2$, $\ddot{\phi} = -2(\dot{\theta}\dot{\phi}/\tan\theta)$, which in the Hamiltonian formulation gives $\hat{L}^2 = -\hbar^2[\partial_\theta^2 + \cot\theta\,\partial_\theta + \csc^2\theta\,\partial_\phi^2]$, the standard angular momentum operator with eigenfunctions Y_ℓ^m. The partition quantum numbers (ℓ, m) are the quantum numbers of this operator: they label the rows of the irreducible representations of $SO(3)$. □

Corollary 2.3 (Angular Basis Completeness)**.** *The set $\{Y_\ell^m(\theta, \phi) : \ell = 0, \ldots, n-1,\ m = -\ell, \ldots, \ell\}$ is a complete orthonormal basis for functions on the sphere S^2 at partition level n.*

3 The Partition Angular Resolution Theorem

Definition 3.1 (Angular Resolution of a Partition Detector)**.** A partition detector at level n can *resolve* two angular directions (θ_1, ϕ_1) and (θ_2, ϕ_2) if and only if the angular separation $\Delta\theta = \arccos(\hat{n}_1 \cdot \hat{n}_2)$ exceeds the spacing between adjacent partition angular modes.

Theorem 3.2 (Partition Angular Resolution Limit)**.** *The minimum resolvable angular separation for a partition detector at level n is:*

$$\Delta\theta_n = \frac{\pi}{2n-1}. \tag{1}$$

For large n: $\Delta\theta_n \approx \pi/(2n)$.

Proof. The angular states at level n are indexed by (ℓ, m) with $\ell \le n-1$. The maximum angular frequency resolvable is $\ell_{\max} = n-1$; the corresponding spherical harmonic Y_{n-1}^{n-1} has $n-1$ oscillation nodes in θ over the range $[0, \pi]$, giving a minimum angular spacing of $\pi/(n-1+1/2) = \pi/(n-1/2) \approx \pi/(2(n-1/2)) = \pi/(2n-1)$, as in the Nyquist criterion for the Legendre polynomial zeros.

Equivalently: the $(2n-1)$ values of m for $\ell = n-1$ divide the azimuthal circle into $2(n-1)+1 = 2n-1$ sectors of width $\Delta\phi = 2\pi/(2n-1)$. The polar resolution matches this: $\Delta\theta = \pi/(2n-1)$. □

Corollary 3.3 (Rayleigh Criterion Equivalence)**.** *Equation (1) is equivalent to the Rayleigh criterion $\Delta\theta = 1.22\lambda/D$ when the partition level n is identified with the ratio D/λ (aperture in wavelength units): $\pi/(2n-1) \approx \pi/(2D/\lambda) = \pi\lambda/(2D)$ (up to the numerical factor $1.22/(\pi/2) = 0.778$, which accounts for the circular vs. rectangular aperture geometry).*

The partition derivation requires no wave optics: it is a pure counting argument on the number of distinguishable angular modes at partition level n.

Remark 3.4 (Why the factor of 1.22)*.* The numerical discrepancy 1.22 vs. $\pi/2 = 1.57$ arises because the Airy pattern is the Fourier transform of a *circular* aperture function (Bessel J_0), while the partition counting uses the $(2n-1)$ modes of the full sphere. The correction factor is $1.22/(\pi/2) = 2 \times 1.22/\pi \approx 0.777$, consistent with the ratio of the first zero of J_0 to the first zero of a square aperture function.

4 Angular Sampling Theorem for Calorimeters

Theorem 4.1 (Angular Sampling Theorem)**.** *A calorimeter with angular cell size $\Delta\eta \times \Delta\phi$ can faithfully reconstruct all angular modes up to:*

$$\ell_{\max} = \frac{\pi}{\Delta\theta} - \frac{1}{2}, \tag{2}$$

where $\Delta\theta = \max(\Delta\eta, \Delta\phi)$ (the coarser of the two granularities). Angular modes with $\ell > \ell_{\max}$ are aliased: high-frequency angular structure maps back to lower ℓ in the reconstructed spectrum.

Proof. By the Nyquist theorem for spherical harmonics: a function on the sphere with bandwidth $\ell_{\max}$ (highest non-zero multipole) can be reconstructed from samples on a uniform grid of spacing $\Delta\theta \le \pi/(2\ell_{\max}+1)$ (the Driscoll-Healy sampling theorem (Driscoll and Healy, 1994)). Solving for $\ell_{\max}$: $\ell_{\max} \le \pi/\Delta\theta - 1/2$. The maximum resolvable ℓ is the floor of this expression. □

Example 4.2 (ATLAS EM calorimeter). The ATLAS EM calorimeter layer 2 has granularity $\Delta\eta \times \Delta\phi = 0.025 \times 0.025$, giving $\Delta\theta \approx 0.025$ rad: $\ell_{\max} = \pi/0.025 - 1/2 \approx 125$. Angular modes $\ell \leq 125$ are faithfully reconstructed. Electromagnetic shower shapes with $\ell > 125$ (e.g., very narrow QCD jets with $\Delta R < 0.025$) are aliased onto lower multipoles.

Example 4.3 (ATLAS HCAL). The ATLAS hadronic calorimeter has $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$, giving $\ell_{\max} \approx 31$. Jet substructure observables with angular structure at $\ell > 31$ require the finer EM calorimeter granularity for resolution.

5 Angular Composition-Inflation

Theorem 5.1 (Angular Composition-Inflation Formula). *The number of distinguishable angular patterns at partition depth n in a d-branching hierarchy is:*

$$A(n,d) = (2n-1) \cdot T(n,d) = (2n-1)\, d\, (d+1)^{n-1}, \quad (3)$$

growing faster than the state count $T(n,d)$ by the factor $(2n-1)$.

Proof. At each partition level n, the $T(n,d)$ kinematic states split into $2n-1$ angular sub-patterns (the $2\ell_{\max}+1 = 2(n-1)+1 = 2n-1$ distinct orientations of the maximum-ℓ mode). Each kinematic state carries one of $(2n-1)$ distinct angular faces, so the total distinguishable patterns are $A(n,d) = (2n-1) \cdot T(n,d)$. □

Corollary 5.2 (Detector information content). *The total information content of a detector at Planck depth n_{P} is: $A(n_{\mathrm{P}},d) = (2n_{\mathrm{P}}-1) \cdot T(n_{\mathrm{P}},d)$. For $n_{\mathrm{P}} = 60$, $d = 3$: $A(60,3) = 119 \times 3 \times 4^{59} \approx 1.2 \times 10^{37}$ distinguishable angular patterns per event. This is the maximum information capacity of the LHC detector.*

6 Angular Resolution and Trigger Design

Theorem 6.1 (Angular Resolution as Trigger Selectivity). *A Level-1 trigger with angular seed threshold $\Delta\theta_{\mathrm{trig}}$ selects all events with angular modes $\ell > \pi/\Delta\theta_{\mathrm{trig}} - 1/2$, and rejects all events whose angular structure lies entirely within modes $\ell \leq \pi/\Delta\theta_{\mathrm{trig}} - 1/2$.*

Proof. The trigger fires when an angular concentration (energy deposit) exceeds a threshold in a cell of size $\Delta\theta_{\mathrm{trig}}$. In spherical harmonic language, this selects for high-ℓ modes (narrow, concentrated depositions correspond to $\ell \gg 1$). By the Angular Sampling Theorem (Theorem 4.1), the cell size $\Delta\theta_{\mathrm{trig}}$ cannot distinguish modes below $\ell_{\mathrm{trig}} = \pi/\Delta\theta_{\mathrm{trig}} - 1/2$. Events with all power at $\ell \leq \ell_{\mathrm{trig}}$ (diffuse, minimum-bias-like events) do not fire the trigger. □

Remark 6.2 (Optimal angular seed size). The minimum useful trigger seed is at $\Delta\theta_{\mathrm{trig}} = \Delta\theta_n$ (the partition resolution limit at Planck depth). A trigger finer than this cannot benefit from further granularity (the angular modes are already fully resolved). This gives the optimal calorimeter seed size $\Delta\theta_{\mathrm{opt}} = \pi/(2n_{\mathrm{P}}-1) = \pi/119 \approx 0.026$ rad, consistent with the ATLAS EM calorimeter design (0.025 rad). The agreement is not a coincidence: the detector was engineered to the same information-theoretic optimum that the partition algebra predicts.

7 Discussion

7.1 Diffraction without waves

The Rayleigh criterion is usually understood as a wave phenomenon: two point sources are resolvable when their Airy patterns overlap no more than the first dark ring of one coincides with the centre of the other. The partition derivation (Theorem 3.2) shows this is more fundamental: it is the impossibility of fitting more than $2n-1$ distinguishable angular modes into the sphere at partition level n. Wave optics implements this counting constraint via the zeros of the Bessel function; the partition algebra implements it via the multiplicity of the $SO(3)$ representation.

7.2 Connection to angular momentum conservation

The Angular Sampling Theorem (Theorem 4.1) is the spatial-domain analog of angular momentum conservation: detecting a pattern at angular mode ℓ conserves $\hbar\sqrt{\ell(\ell+1)}$ of angular momentum. The Nyquist constraint on $\ell_{\max}$ is the constraint that no more angular momentum is attributed to an event than the detector's calorimeter can spatially resolve.

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